

GATE

ALL BRANCHES



General Aptitude

QUANTITATIVE APTITUDE

Lecture No.- 14



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Recap of Previous Lecture



Topic 1

Trains

(Time & Distance)



Topics to be Covered



Topic-1

Boats Streams & Races

Topic-2

Mensuration 2 D & 3D



[MCQ]



#Q. If a train travelling at 40 km/hr crosses another train of length 100 meter travelling at 14 km/hr in opposite directions in 30 seconds, then find the length of the train. (in meters)

Assignment

$$D = S \times T$$

$$l_{T_1} + 100 = 15 \times 30$$

$$\Rightarrow l_{T_1} = 450 - 100$$

$$l_{T_1} = 350 \text{ m}$$

$$D = \underline{l_{T_1}} + 100$$

$$R.S. = \frac{3}{54} \text{ km/hr} \times \frac{5}{18} \left(\frac{\text{m}}{\text{sec}} \right) = 15 \text{ m/sec}$$

$$T = 30 \text{ seconds}$$

[MCQ]

$$D = S \times T$$



#Q. A train running at 52 km/hr takes 36 seconds to pass a platform. Next it takes 24 seconds to cross a man walking at the platform with 10 km/hr in the same direction. Find the length of the platform. (in meters)

Assignment

Train Cross Platform

$$D = l_T + l_P$$
$$= 280m + l_P$$

$$S = 52 \text{ km/hr} \left(\times \frac{5}{18} \right) \text{ m/sec}$$

$$T = 36 \text{ sec}$$

$$\Rightarrow 280 + l_P = 52 \times \frac{5}{18} \times 36$$

$$\Rightarrow l_P = 520 - 280 = \underline{\underline{240m}}$$

Train cross a man

$$D = l \text{ of Train}$$

$$R.S = 42 \text{ km/hr} \times \left(\frac{5}{18} \right) \text{ (m/sec)}$$

$$T = 24 \text{ sec}$$

$$l \text{ of Train} = 42 \times \frac{5}{18} \times 24 = \underline{\underline{280m}}$$

Boats & Streams



$$(US) \text{ Upstream Speed } \boxed{US = S.B - S.W}$$

$$(DS) \text{ Downstream Speed } \boxed{DS = S.B + S.W}$$



$\Rightarrow 8 \text{ km/hr}$

! (DS) Downstream Speed

$$S.B = \frac{DS + US}{2}$$

$$S.W = \frac{DS - US}{2}$$

10 km/hr

6 km/hr



[MCQ]



#Q. A man rows his boat downstream @ 18 km/hr & upstream @ 10 km/hr.

Find the speed of boat in still water.

$$S.B = \frac{DS + US}{2} = \frac{18 + 10}{2} = \frac{28}{2} = 14 \text{ km/hr}$$

A 12 km/hr

B 10 km/hr

C 14 km/hr

D 16 km/hr



[MCQ]



$$T = \frac{\text{Dist}}{\text{Speed}}$$

#Q. A boat traveled from A to B and back to A from B in 5 hours. If the speed of boat in still water and the speed of stream be 7.5 kmph and 1.5 kmph, then what is the distance between A and B? = 'x' km

A 18 km

B 10 km

C 16 km

D 5 km

$$\frac{x}{9} + \frac{x}{6} = 5$$

$$\Rightarrow \frac{5x}{18} = 5$$

$$\therefore x = 18 \text{ km}$$

Races

$$\frac{A}{B} \times \frac{B}{C} = \frac{1000}{900} \times \frac{1000}{950} = \frac{1000}{855}$$

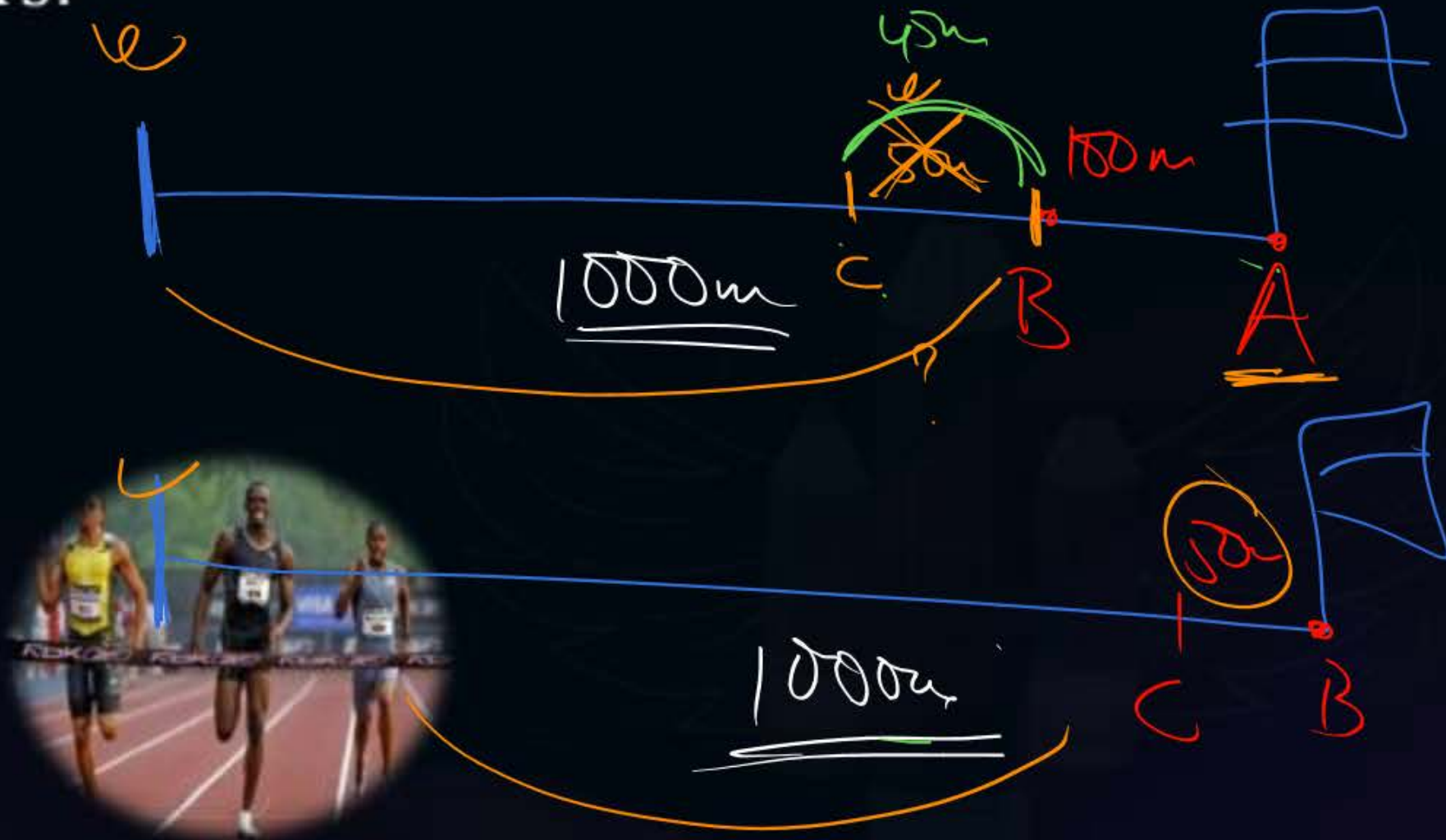
145m



#Q. In a km race A defeats B by 100 meters and B defeats C by 50 meters, then A defeats C by how many meters?

$$\frac{A}{B} = \frac{1000}{900}$$

$$\frac{B}{C} = \frac{1000}{950}$$



[MCQ]



#Q. In a km race A defeats (beats) B by 200 meters and B defeats C by 100 meters, then A defeats C by how many meters?

$$\frac{A}{C} = \frac{1000}{800} \times \frac{1000}{900}$$

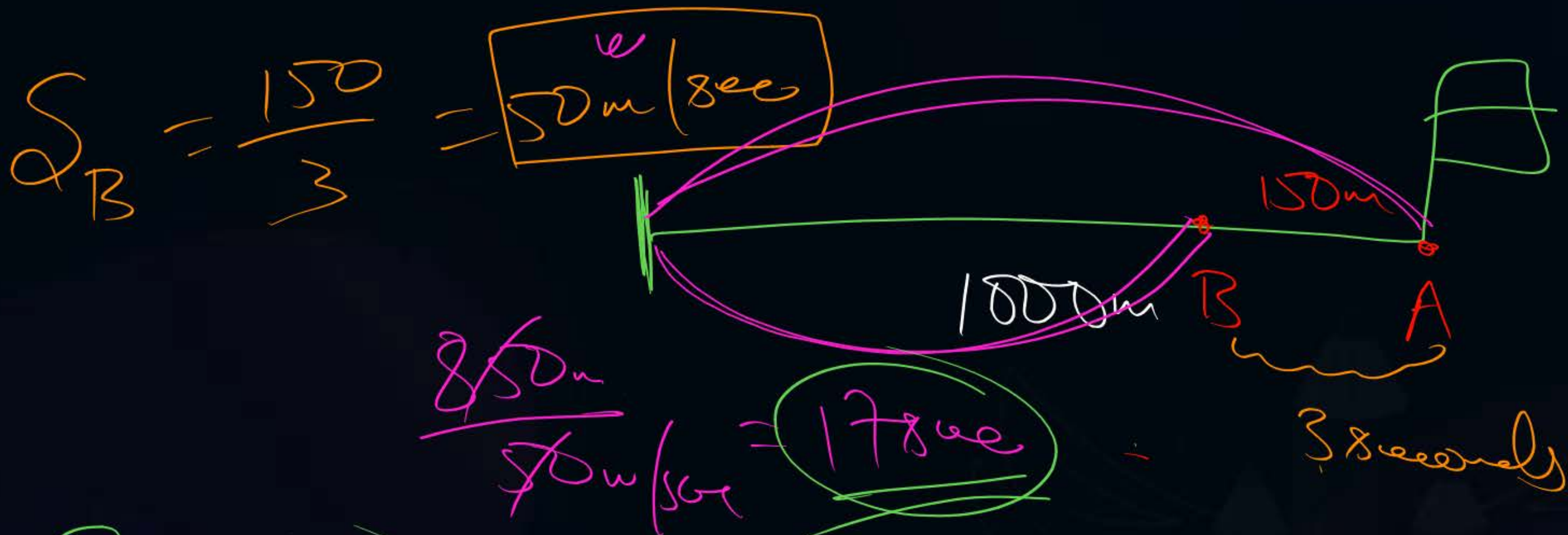
$$\frac{A}{C} = \frac{1000}{720}$$

280m

[MCQ]



#Q. In a km race A defeats B by 150 meters or 3 seconds. Find the speed of A.



$$S_A = \frac{1000}{17} = 58.82 \text{ m/sec}$$

[MCQ]

$$S_A = ? \text{ 50m/sec}$$

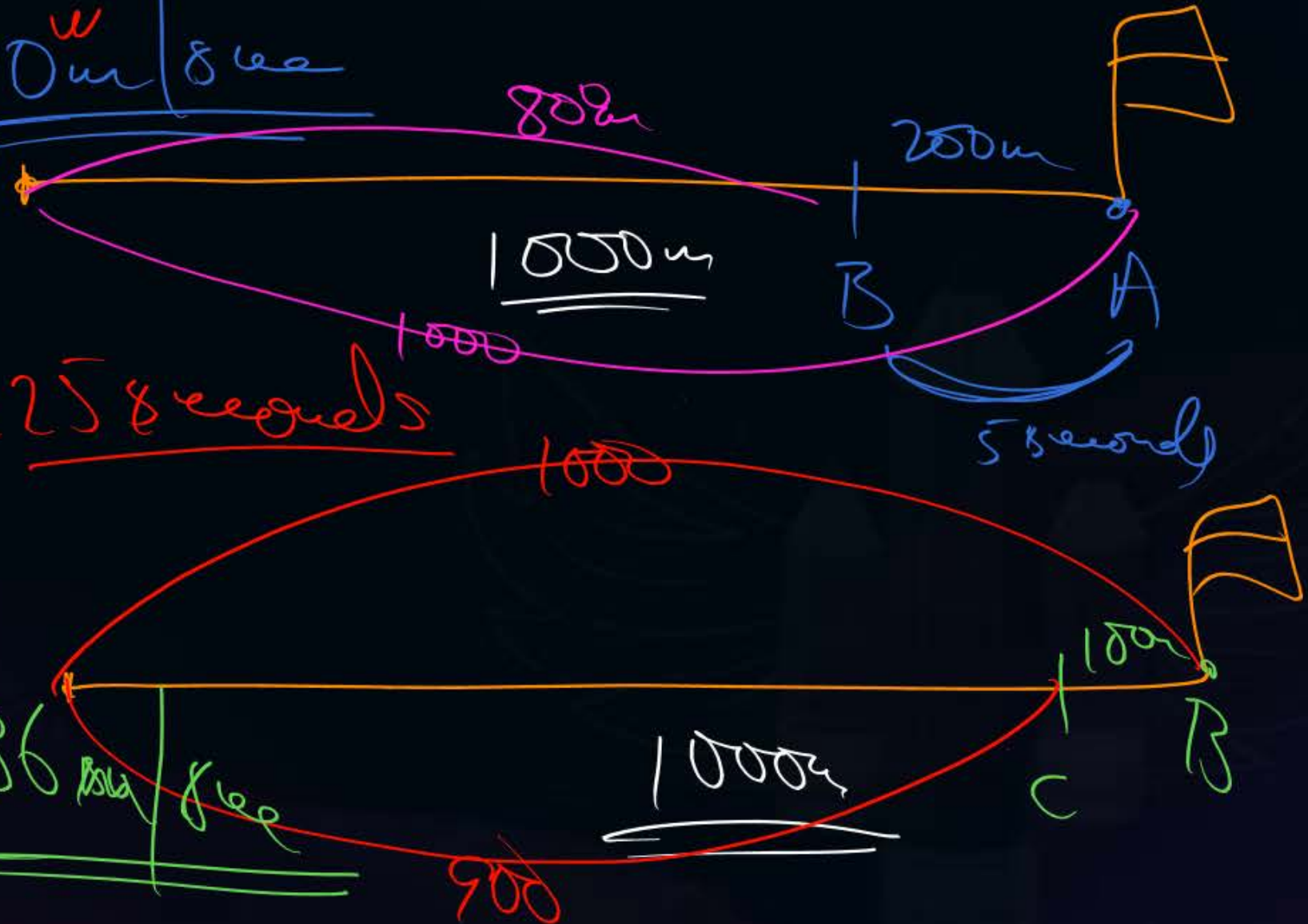


#Q. In a km race A defeats B by 200 meters or 5 seconds whereas B defeats C by 100m. Find the speed of C. (m/sec)

$$S_B = \frac{200}{5} = 40 \text{ m/sec}$$

$$\frac{1000}{40} = 25 \text{ seconds}$$

$$S_C = \frac{900}{25} = 36 \text{ m/sec}$$

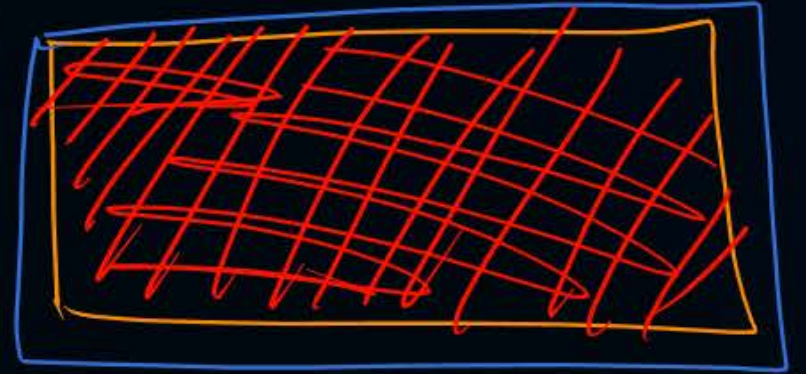


MENSURATION: 2D



Perimeter (P)

Area (A)



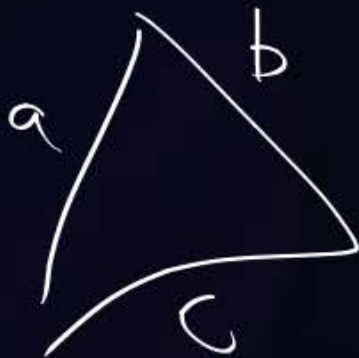
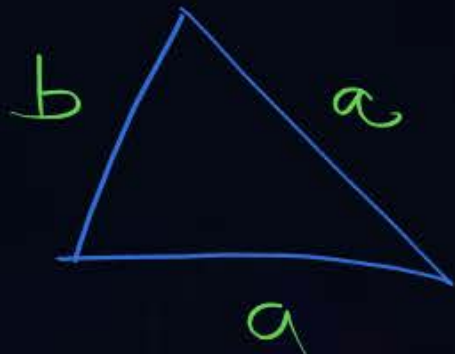
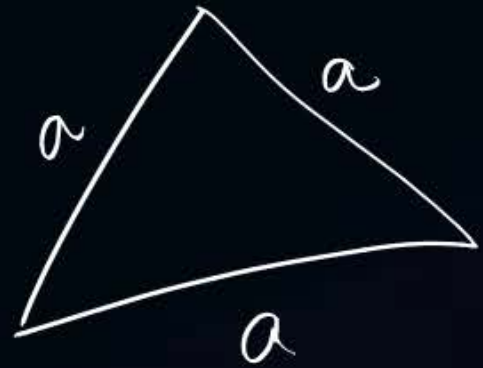
Triangle

Quadrilateral

Circles

Triangles:

(Types)



I $\Rightarrow$ (Side)

✓ 1. Equilateral $\triangle$

✓ 2. Isosceles $\triangle$

3. Scalene $\triangle$

(Angles)

II $\Rightarrow$

1. Right $\triangle$

2. Acute $\triangle$

3. Obtuse $\triangle$



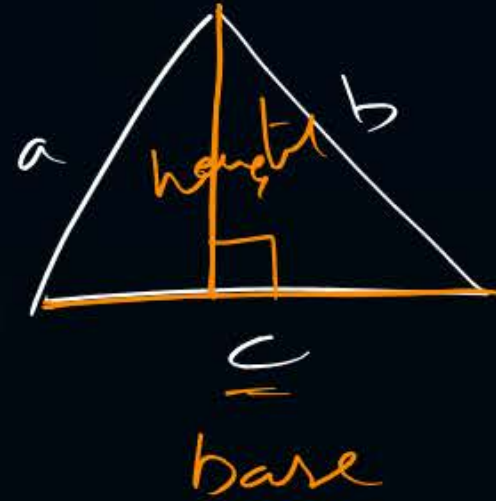
Triangles:



Heron's Formulae

$$P = a + b + c$$

$$A = \sqrt{s(s-a)(s-b)(s-c)}$$



$$\text{where } s = \frac{a+b+c}{2}$$

$$A = \frac{1}{2} \times b \times h$$

Area of equilateral Δ

$$= \frac{\sqrt{3}}{4} a^2$$

Quadrilaterals:

$$P = AB + BC + CD + DA$$

$$A = \left(\frac{1}{2} \times d \times h_1 \right) + \left(\frac{1}{2} \times d \times h_2 \right)$$

$$= \frac{1}{2} d (h_1 + h_2)$$

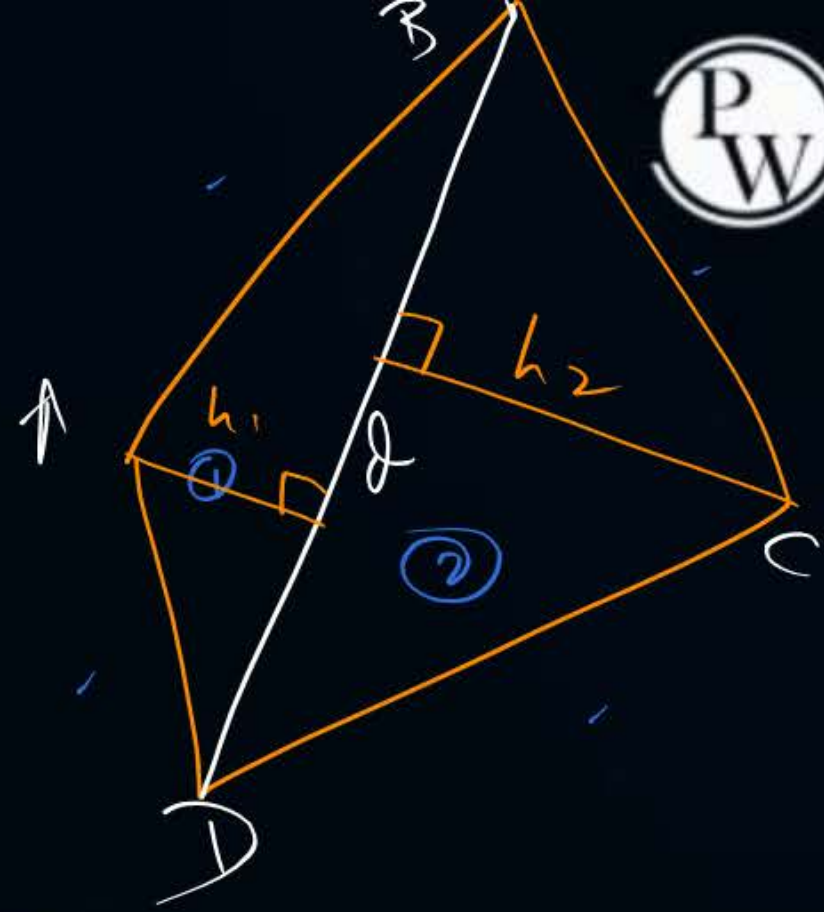
Rectangle

Square

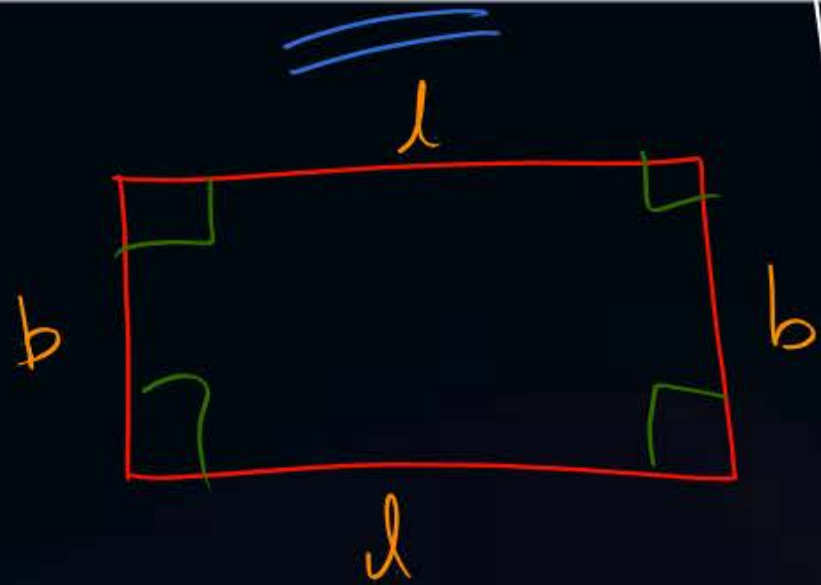
Parallelogram

Trapezium

etc



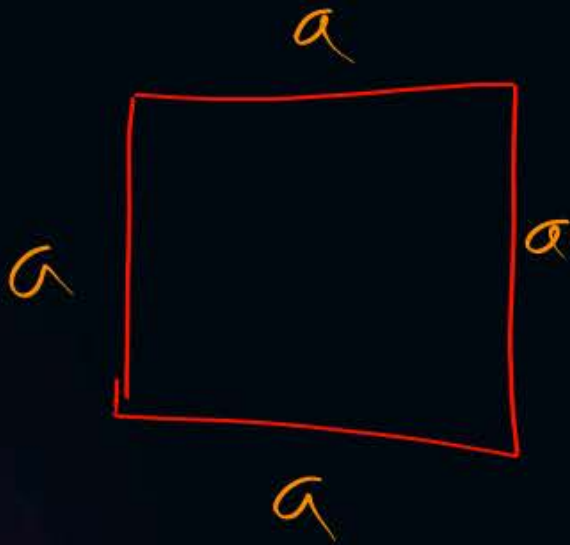
Rectangle and Square:



$$P = 2l + 2b$$

$$= 2(l + b)$$

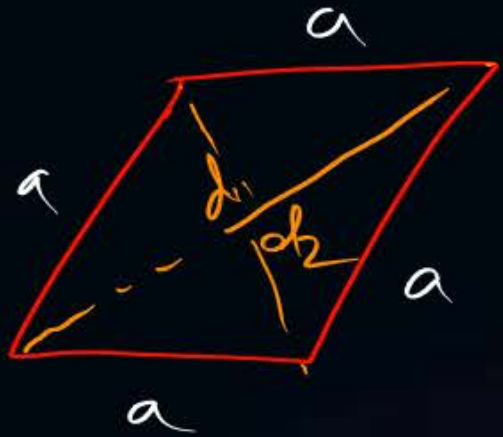
$$A = l \times b$$



$$P = 4a$$

$$A = a^2$$

Rhombus, Trapezium & Parallelogram:



$$P = 4a$$

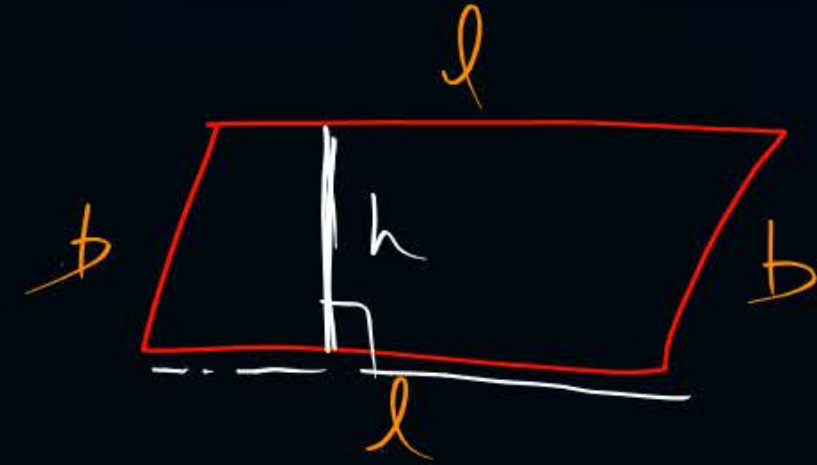
$$A = \frac{1}{2} \times d_1 \times d_2$$



$$P = a + b + c + d$$

$$A = \frac{1}{2} \times (\text{sum of parallel sides}) \times \text{height}$$

$$= \frac{1}{2} \times (a + c) \times h$$



$$P = 2(l + b)$$

$$A = \text{base} \times \text{height}$$

$$(l \times h)$$

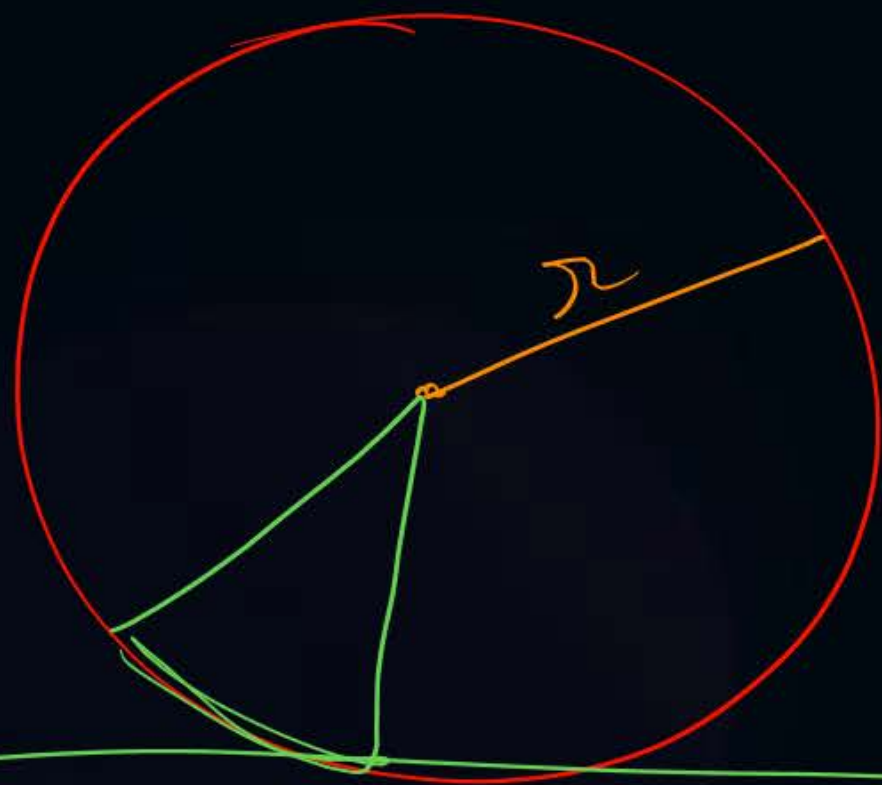
Circles & Sectors:



~~C~~ = Circumference
(C)

r = radius

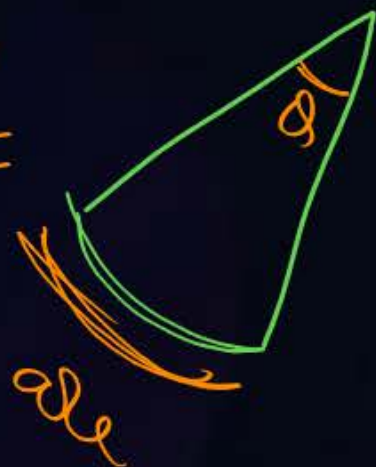
$$\pi = \frac{22}{7} \text{ OR } 3.14$$



$$C = 2\pi r$$

$$A = \pi r^2$$

Sector



$$\text{Length of Arc} = \frac{\theta}{360} \times 2\pi r$$

$$\text{Area of sector} = \frac{\theta}{360} \times \pi r^2$$

MENSURATION: 3D



Lateral Surface Area
(LSA)

Total Surface Area
(TSA)

Volume (V)

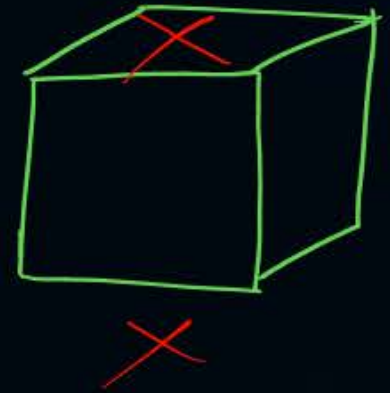
Cube

Cuboid

Cone

Cylinder

Sphere etc



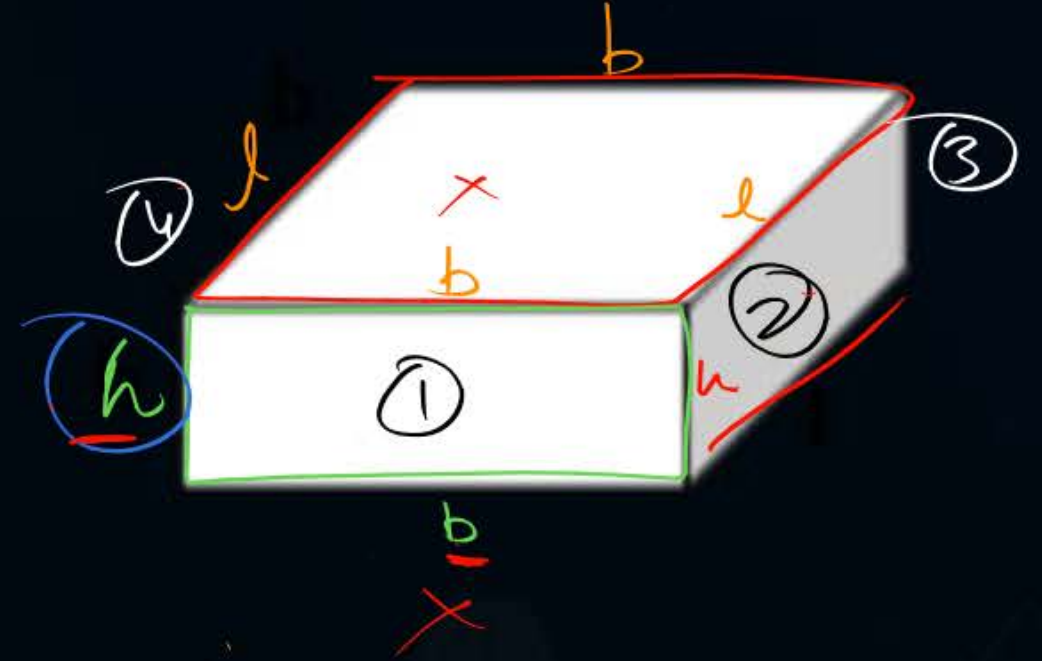
Cuboid:



$$V = \underline{l \times b \times h}$$

$$LSA = 2bh + 2lh$$
$$= 2h(l+b)$$

$$TSA = 2bh + 2lh + 2lb$$
$$= 2(lb + bh + hl)$$



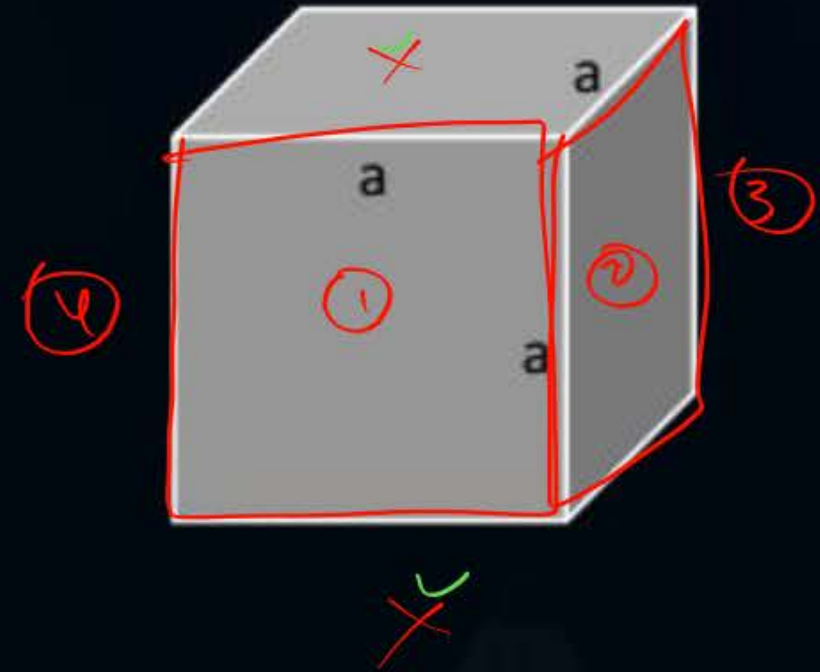
Cube:



$$V = a^3$$

$$LSA = 4a^2$$

$$TSA = 6a^2$$



Cylinder:

~~LSA~~ = Curved Surface Area (CSA)

$$V = \pi r^2 h$$

$$CSA = 2\pi r h$$

$$\begin{aligned} TSA &= 2\pi r h + 2\pi r^2 \\ &= 2\pi r(h + r) \end{aligned}$$



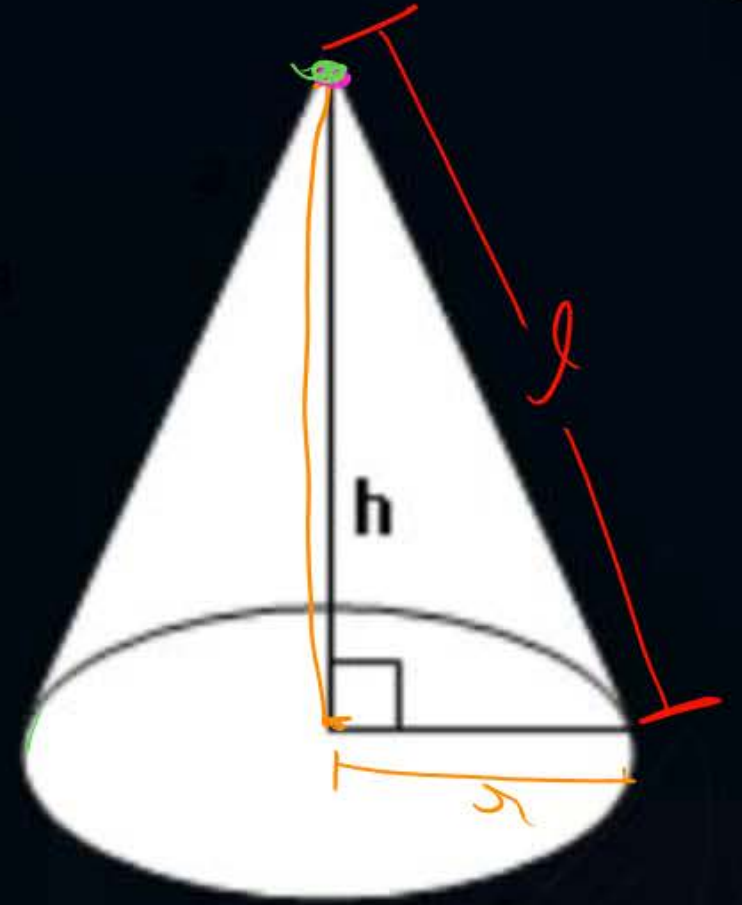
Cone:



$$V = \frac{1}{3} \pi r^2 h$$

$$CSA = \pi r \underline{l}$$

$$\begin{aligned} TSA &= \pi r l + \pi r^2 \\ &= \pi r (l + r) \end{aligned}$$



Sphere:

$$V = \frac{4}{3} \pi r^3$$

$$TSA = 4 \pi r^2$$



Hemi-sphere:

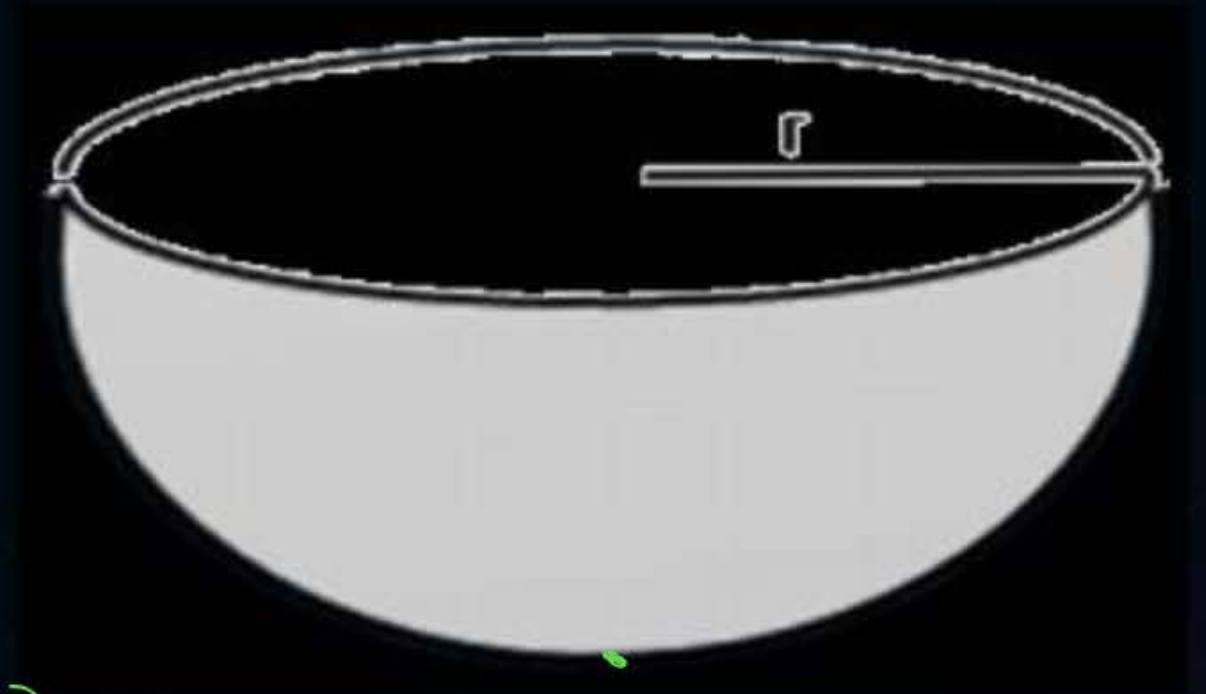
$$V = \frac{1}{2} \times \frac{4}{3} \pi r^3$$

$$= \frac{2}{3} \pi r^3$$

$$CSA = 2\pi r^2$$

$$TSA = 2\pi r^2 + \pi r^2$$

$$= 3\pi r^2$$



[MCQ]



#Q. What would be the area of sector of a circle whose radius is 12 cm and the length of the arc is 20 cm?

Assignment

- A** 60 sq. cm
- B** 240 sq. cm
- C** 120 sq. cm
- D** 64 sq. cm

[MCQ]



#Q. The cross-section of a canal is a trapezium in shape. If the canal is 7 m wide at the top and 9 m at the bottom and the area of cross-section is 128 sq m, find the height of the canal.

Assignment

- A** 32 m
- B** 8 m
- C** 16 m
- D** 4 m

[MCQ]



#Q. A cube of 6 cm side melted and smaller cubes of 2 cm side are manufactured.
Find the number of smaller cubes so formed.

Assignment

- A** 12
- B** 27
- C** 24
- D** 8

[MCQ]



#Q. The length, breadth & height of a cuboid are in the ratio of 4 : 3 : 2 and its volume is 3000 m^3 . Find its total surface area.

Assignment

- A** 1300 m^2
- B** 1500 m^2
- C** 1333 m^2
- D** 2700 m^2

[MCQ]



#Q. The dimensions of a room are 20 m x 15 m x 10 m. What is the cost of painting its four walls at the rate of ₹ 15 per 100 sq m?

Assignment

- A** ₹ 70
- B** ₹ 105
- C** ₹ 225
- D** None of these

[MCQ]



#Q. In two concentric circles, a chord length 80 cm of larger circle becomes a tangent to the smaller circle whose radius is 9 cm. The radius of the larger circle will be?

Assignment

- A** 13 cm
- B** 41 cm
- C** 52 cm
- D** 75 cm

[MCQ]



#Q. The diagonal of a rectangle is $\sqrt{41}$ cm and its area is 20 sq cm. The perimeter of the rectangle is?

Assignment

- A** 9 cm
- B** 18 cm
- C** 20 cm
- D** 41 cm



2 mins Summary



Topic

Mensuration 2D & 3D





THANK - YOU